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Health Psychology Review

ISSN: 1743-7199 (Print) 1743-7202 (Online) Journal homepage: www.tandfonline.com/journals/rhpr20

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To cite this article: Dominika Kwasnicka, Justin Presseau, Martin White & Falko F. Sniehotta (2013) Does planning how to cope with anticipated barriers facilitate health-related behaviour change? A systematic review, Health Psychology Review, 7:2, 129-145, DOI: 10.1080/17437199.2013.766832

To link to this article: https://doi.org/10.1080/17437199.2013.766832

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Published online: 19 Feb 2013.

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Health Psychology Review, 2013
Vol. 7, No. 2, 129–145, http://dx.doi.org/10.1080/17437199.2013.766832

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Does planning how to cope with anticipated barriers facilitate health-related behaviour change? A systematic review

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(*Received 23 September 2012; final version received 23 December 2012*)

Health-related interventions often prompt participants to plan how to cope with anticipated barriers to behaviour change, a technique known as coping planning. The purpose of this study was to review the evidence of the efficacy of prompting individuals to form coping plans as a technique for promoting health-related behaviour change. Electronic databases (MEDLINE, Embase and PsycInfo) and unpublished literature were searched for randomised controlled trials that allocated participants to the study conditions with and without prompts to form coping plans. Evidence was assessed for quality and narratively synthesised. Full text papers of 65 articles were assessed for eligibility and 11 papers were included in the review. Coping planning interventions appear to be efficacious when participants are supported in the process of forming coping plans. Combining action plans with coping plans seems to be more efficacious than using action plans only. The overall efficacy of coping planning is variable. Future interventions should consider potential moderators of the efficacy of such plans.

Keywords: barriers; behaviour change; coping; implementation intentions; planning

People often intend to change their health behaviour but subsequently fail to take action; a discrepancy known as the intention–behaviour gap (Orbell & Sheeran, 1998). Action planning (i.e., specifying when, where and how to perform a behaviour) (Gollwitzer, 1999) and coping planning (i.e., specifying how to deal with anticipated barriers) (Sniehotta, Schwarzer, Scholz, & Schuz, 2005) are two behaviour change techniques (Michie & Johnston, 2012) that focus on helping motivated individuals to bridge the gap between their intention and behaviour. Action plans differ from coping plans in their function: the former are used to facilitate actions and the latter to inhibit distractions (Sniehotta, 2009).

The beneficial effects of action planning have been demonstrated for a number of health-related behaviours: smoking cessation (Armitage, 2007; Conner & Higgins, 2010), healthy eating (Verplanken & Faes, 1999), exercising (Sniehotta, Scholz, & Schwarzer, 2005), oral hygiene (Sniehotta, Soares, & Dombrowski, 2007), breast self-examination (Orbell, Hodgkins, & Sheeran, 1997), stair use (Kwak, Kremers, van Baak, & Brug, 2007) and taking a vitamin supplements (Sheeran & Orbell, 1999). Previous reviews have demonstrated the efficacy of action plans for behaviour change (Gollwitzer & Sheeran, 2006) and showed that interventions involving behaviour

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change techniques aimed at barrier management (such as relapse prevention) are more efficacious than other interventions (Dombrowski et al., 2012) and suggested that planning interventions incorporating barrier management are more effective in changing physical activity behaviour than simple action planning interventions (Bélanger-Gravel, Godin & Amireault, 2011). To date, however, no systematic review has yet assessed the efficacy of coping planning as such.

The aim of this study was to systematically review the literature of randomised controlled trials (RCTs) that test the efficacy of coping planning as a separate component of health-related behaviour change interventions.

Methods

The systematic review protocol was registered in PROSPERO – an international *database* of prospective systematic reviews in health and social care (protocol registration number CRD42011001272).

Search strategy

MEDLINE, Embase and PsycInfo databases were searched using a comprehensive search strategy (see Supplemental Material for additional details of the search). The search period ranged between 1 January 2004 [when a relevant published systematic review closed its search period (Gollwitzer & Sheeran, 2006)] and 1 March 2012. Search terms included were: 'coping', 'barrier(s)', 'obstacle(s)', 'shielding', 'implementation intention(s)', 'plan(s/ing)', 'behaviour(s)' as well as appropriate synonyms, adjusted in accordance with the particular database. Moreover, all trials included in Gollwitzer and Sheeran's (2006) systematic review of planning interventions were assessed for inclusion. Reference lists of relevant review articles and of all included papers were searched manually for further studies meeting the inclusion criteria. A search of grey literature was undertaken, consisting of a call for unpublished trials and papers sent to mailing lists of international health psychology and behavioural medicine societies.

Eligibility criteria

Included papers had a title and abstract in English and used coping planning alone or as a part of a health-related behaviour change intervention. 'Coping planning' was defined as planning how to overcome barriers which may occur during health-related behaviour change.

Participants. Only studies which employed non-vulnerable adult participants (with a mean/median age of $\geq$18 years) were included. There were no limitations regarding participants' background, education level or pre-diagnosed conditions.

Interventions. Studies had to include participants randomised to a group that was explicitly asked to form plans on how to cope with barriers which may occur during any health-related behaviour change.

Comparisons. Included studies were required to contain a comparator group, which although provided similar content to that used in the coping planning intervention, excluded coping planning.

Outcomes. Included studies were required to report a health-related behaviour as an outcome measure(s) of the behaviour intervention.

Study designs. Studies reported as RCTs were included in the review.

Screening procedure

One reviewer (DK) screened all studies identified through initial searches against eligibility criteria in two stages. Initial screening was based on screening titles, abstracts and keywords. Then full papers of potentially relevant articles were retrieved and reviewed against eligibility criteria by two researchers (DK and FFS).

Data extraction

The data extraction form was combined with a risk of bias assessment tool and it followed Cochrane Collaboration handbook guidance (Higgins & Green, 2011). If any study-related information was missing or unclear, the authors of these studies were approached and additional study details were requested. Data were independently extracted by two researchers (DK and JP).

The data extraction procedure was divided into three stages: (1) general data extraction from papers using a data extraction form to identify study characteristics including source, methods, participants, interventions, outcomes, results, funding sources, (2) assessing the quality of the included studies using the Cochrane risk of bias assessment tool and (3) identifying comparable conditions, which included coping plans, action plans or a combination of both across included studies, and analysing comparable data-sets and measures. Descriptive information was presented to provide a qualitative summary of all included studies.

Results

A total of 349 potential articles were first identified. Twenty-four additional reports from other sources were screened for inclusion. After exclusion of duplicates and articles not meeting the inclusion criteria, 65 articles were retrieved and full-text articles were assessed for eligibility. Of those, 23 articles were included from Gollwitzer and Sheeran's (2006) systematic review. Eleven articles met all inclusion criteria (see Figure 1). All the included studies were described as RCTs although the randomisation procedure was not always clear (see Table 1). Thus, we labelled some of the included studies as quasi-RCTs and we defined them as studies reported by authors as an RCT, but with some degree of uncertainty concerning bias in the randomisation procedure.

Study characteristics

The included studies' sample sizes varied from 44 to 2082 participants. Studies included people in cardiac rehabilitation ($n = 2$), students ($n = 3$), smokers ($n = 2$),

sedentary people with a spinal cord injury (SCI) ($n = 1$), older women ($n = 1$), people at a lower socio-economic status ($n = 1$) and a general population sample ($n = 1$). Studies assessed a variety of health-related behaviours including increasing physical activity ($n = 5$), moderating binge drinking ($n = 2$), preventing relapse of smoking ($n = 2$), increasing uptake of mammography ($n = 1$) and reducing alcohol consumption ($n = 1$).

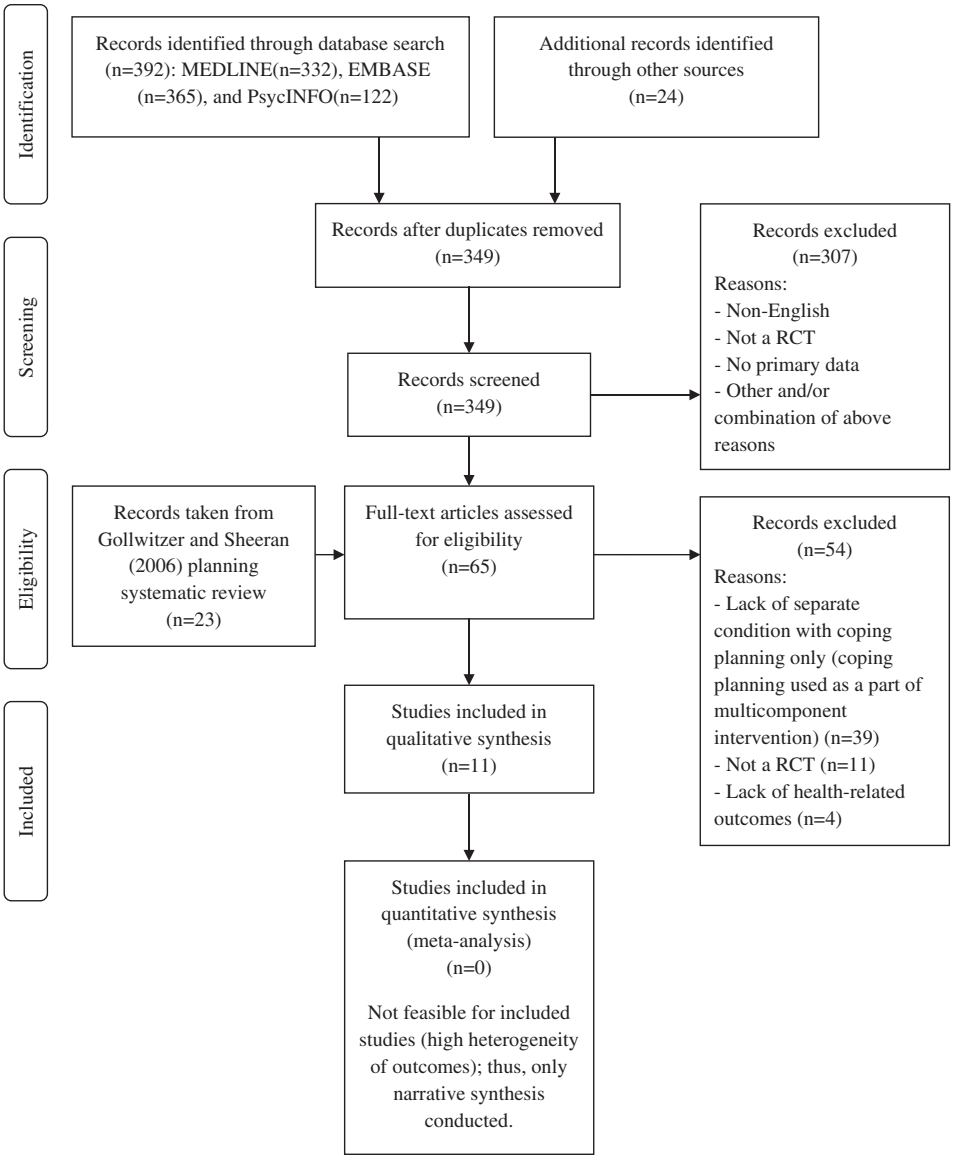

Figure 1. Results for search strategy and screening procedure.

Risk of bias assessment

A risk of bias assessment was conducted by two study authors (DK and JP) following the process detailed in the Cochrane Collaboration handbook (Higgins & Green, 2011) and high agreement was reached between reviewers (Krippendorff's $\alpha = 0.70$). The risk of bias assessment suggested that not all included studies used appropriate random allocation procedures. One study used page numbering as an indicator for condition allocation (Rutter, Steadman, & Quine, 2006) and two studies allocated subsequently approached participants alternately into different study conditions (Scholz, Sniehotta, Burkert, & Schwarzer, 2007; Sniehotta, Scholz, & Schwarzer, 2006).

Some of the included studies reported high attrition rates – over 20% of participants (Skar, Sniehotta, Molloy, Prestwich, & Araujo-Soares, 2011; van Osch, Lechner, Reubsaet, Wigger, & De Vries, 2008). Furthermore, some of the included studies used participants' self-reported outcomes, where more objective measures could have been used, for example, for physical activity level measurements (Arbour-Nicitopoulos, Ginis, & Latimer, 2009; Scholz et al., 2007; Sniehotta et al., 2006) or for smoking status assessment (Armitage, 2008; van Osch et al., 2008). The results of the risk of bias assessment tool suggested moderate quality within and between included studies.

Coping plans formulation

Four studies prompted participants to form plans for overcoming self-identified barriers and seven studies pre-specified these barriers (see Table 1). The number of coping plans that participants were asked to form differed between included studies and ranged from one to three plans for self-identified barriers. When participants were asked to match pre-specified barriers with plans to overcome them, usually 20 situational scenarios were provided ($n = 4$). In an intervention promoting mammography uptake, participants were prompted to form coping plans for three pre-specified barriers to behaviour change (Rutter et al., 2006). In an intervention to prevent smoking relapse, participants were presented with 14 behavioural barriers, asked to choose three personally relevant barriers, then prompted to form plans for overcoming these chosen barriers (van Osch et al., 2008). Thus, in the included studies the majority of behavioural barriers were pre-identified by experiment providers.

Mode of delivery

Included studies also varied in the mode of delivery. Eight used a pen-and-paper format, two were delivered via the Internet (Skar et al., 2011; van Osch et al., 2008) and one intervention was delivered by telephone then formulated plans were recorded by the experimenter and posted to the participants (Arbour-Nicitopoulos et al., 2009). In eight interventions, coping plan formation was monitored and assessed by the experimenter (Adriaanse et al., 2010; Arbour-Nicitopoulos et al., 2009; Arden & Armitage, 2012; Armitage, 2008; Armitage & Arden, 2010; Murgraff, White, & Phillips, 1996; Scholz et al., 2007; Sniehotta et al., 2006). In the remaining three studies (Rutter et al., 2006; Skar et al., 2011; van Osch et al., 2008), coping plans formulated by the participants were not assessed or checked by the intervention providers.

Table 1. Study characteristics.

References/ data-set	Sample characteristics	Behavioural goal	Follow-up period	Study design	Coping plans format, number of plans, structure	Mode of delivery
Arbour-Nicitopoulos et al. (2009)	44 sedentary adults with spinal cord injury (SCI); mean age: 49.7y (SD = 12.71) (32% ♀), Canada	Increasing leisure-time physical activity (LTPA)	10 weeks	AP + CP versus AP	Participants prompted to form plans for overcoming three self-identified behaviour change barriers	Delivered over the phone, recorded by the experimenter and then sent to the participants
Arden and Armitage (2012)	56 students; mean age: 20.57y (SD = 1.90) (66.1% ♀), UK	Moderating binge drinking	Two weeks	CP versus control versus active control	Participants provided with a volitional help sheet which pre-specified 20 possible behaviour change barriers and 20 possible ways of overcoming them	Pen and paper form. Plans pre-specified in the written form
Armitage (2008)	193 smokers; mean age: 37y (SD = 14.6) (51.81% ♀), UK	Smoking cessation	Two weeks	CP versus control versus active control versus control	Participants provided with a volitional help sheet which pre-specified 20 possible behaviour change barriers and 20 possible ways of overcoming them	Pen and paper form. Plans pre-specified in the written form
Armitage and Arden (2012)	69 general population; mean age: 38.51 (SD = 16.34) (52.2%♀), UK	Reducing alcohol consumption	Three months	CP (multiple) versus CP (single) versus IMP (could be CP or AP) versus active control	Participants provided with a volitional help sheet which pre-specified 20 possible behaviour change barriers and 20 possible ways of overcoming them	Pen and paper form. Plans pre-specified in the written form (in CP multiple and single condition)

Table 1 (*Continued*)

References/ data-set	Sample characteristics	Behavioural goal	Follow-up period	Study design	Coping plans format, number of plans, structure	Mode of delivery
Armitage and Arden (2010)	68 people with low socio-economic status (SES); mean age: 27.5y (SD = 12.71) (48.5% ♀), UK	Increasing physical activity	One month	CP versus control	Participants provided with a volitional help sheet which pre-specified 20 possible behaviour change barriers and 20 possible ways of overcoming them	Pen and paper form. Plans pre-specified in the written form
Murgraff et al. (1996)	102 students; mean age:26y (73.5% ♀), UK	Moderating binge drinking	Two weeks	CP versus control	Participants received an option menu of possible responses for refusing a drink, asked to choose one technique and specify a time and place in which the chosen technique would be implemented	Pen and paper form. Plans pre-specified in the written form
Rutter et al. (2006)[a]	2082 older women; mean age: 56.1y (100% ♀), UK	Increasing uptake of mammography	One year max.	CP versus control	Participants prompted to form coping plans for 3 pre-specified behaviour change barriers, in some of them solutions given (select from list), others open ended questions	Pen and paper form. Plans/plans' formulation not assessed
Scholz et al. (2007)[a]	237 cardiac rehabilitation patients; mean age: 59.6y (SD = 9.62) (20.5% ♀), Germany	Increasing physical activity	Ten weeks	AP + CP versus AP versus control	Participants prompted to form plans for overcoming up to three self-identified barriers of behaviour change	Pen and paper form. Facilitated by the interviewer

Table 1 (*Continued*)

References/ data-set	Sample characteristics	Behavioural goal	Follow-up period	Study design	Coping plans format, number of plans, structure	Mode of delivery
Skar et al. (2011)	1273 students; Mean age: 22.8y (SD = 6.7) (63.4% ♀), UK	Increasing physical activity	Eight weeks	AP + CP versus AP versus CP versus control	Participants prompted to form plans for overcoming up to three self-identified barriers of behaviour change	Computer generated. Plans/plans' formulation not assessed
Sniehotta et al. (2006)[a]	246 cardiac rehabilitation patients; mean age: 59.3y (SD = 10) (22% ♀), Germany	Increasing physical activity	Ten weeks	AP + CP versus AP versus control	Participants prompted to form plans for overcoming up to three self-identified barriers of behaviour change	Pen and paper form. Facilitated by the interviewer
van Osch et al. (2008)[a]	1566 smokers; mean age: 36.2y (SD = 11.5) (60.8% ♀), Netherlands	Smoking cessation; relapse prevention	Seven months	CP versus control	Participants presented with 14 behavioural barriers and asked to choose three personally relevant ones, then prompted to form plans for overcoming these barriers	Computer generated. Plans/plans' formulation not assessed

Notes: [a]Randomisation procedure not clear.
AP, action plans; CP, coping plans; AP + CP, intervention condition which combines both forms of planning; RCT, randomised control trial; SD, standard deviation.

Combining coping planning with action planning

Four out of 11 interventions included in the review compared a condition which combined action planning with coping planning and to a condition which involved only action planning (Adriaanse et al., 2010; Arbour-Nicitopoulos et al., 2009; Scholz et al., 2007; Skar et al., 2011; Sniehotta et al., 2006). One of these four studies did not show significant improvement in behaviour when action planning and coping planning were combined and then compared with action planning alone (Skar et al., 2011). The other three interventions which compared a combined action and coping planning condition with action planning alone demonstrated that the combination of action and coping planning was more efficacious than action planning alone (see Table 2).

Coping planning compared to the control-only condition

Eight interventions out of eleven randomly allocated participants to a coping planning or control conditions (see Table 3). Only one study used a 2×2 full factorial design randomly allocating participants to receive prompts to form action plans or not and to form coping plans or not (Skar et al., 2011). Five interventions used a 'volitional help sheet'. In this approach, participants are presented with a list of pre-specified behavioural barriers (i.e., if component) and a list of used pre-specified actions (i.e., then component) and are prompted to form plans by combining 'if' and 'then' components from both lists simply by drawing a line between both components. All five trials of volitional help sheets found this intervention to be efficacious.

Table 2. Efficacy of action plans combined with coping plans versus action plans only.

References/ data-set	Efficacy of AP + CP versus AP only. Outcome measures used (SD) and statistical significance of comparison	Behavioural outcome reported	*N*	Effect size
Arbour-Nicitopoulos et al. (2009)*	After 10 weeks: AP + CP: ($M = 97.83$, SD = 61.11). AP: ($M = 54.51$, SD = 61.11) $p < 0.05$	Leisure-time physical activity (LTPA) reported in min/week	44	$d = 0.71$
Scholz et al. (2007)*	After 10 weeks: AP + CP: ($M = 1.2$, SD = .47). AP: ($M = 0.96$, SD = 0.33) $p < 0.05$	Physical activity reported in average frequency of physical exercise per week	237	$d = 0.59$
Skar et al. (2011)	After eight weeks: AP + CP: ($M = 2.98$, SD = 2.35). AP: ($M = 2.96$, SD = 2.26) $p > 0.05$*	Physical activity reported in average frequency of physical exercise per week	1,273	$d = 0.0086$
Sniehotta et al. (2006)*	Activity 1 (exercise): AP + CP: ($M = 178.94$, SD = 114.33). AP: ($M = 113.26$, SD = 114.37) $p < 0.01$	Physical activity reported in min/week	246	$d = 0.57$

Notes: AP, action plans; CP, coping plans; AP + CP, intervention condition which combines both forms of planning.
*AP + CP condition did statistically significantly outperform AP condition. Data may differ from that reported in the published papers because we prioritised data provided to us by study authors.

Table 3. Efficacy of coping plans versus control comparison.

References/data-set	Efficacy of CP versus Control	Behavioural outcome reported	N	Effect size
Arden and Armitage, (2012)*	After intervention (ITT analysis): CP: ($M = 1.23$, SD $= 0.80$); control: $M = 1.90$, SD $= 0.80$); active control: ($M = 1.82$, SD $= 0.80$); $p = 0.01$.	There were significant decreases in units of alcohol consumed and self-reported binge drinking frequency in the CP condition, but not in either of the control conditions	56	$d = 0.79$
Armitage (2008)*	After the intervention nicotine dependence: $F(3, 192) = 10.84$, $p < 0.01$, $\eta_p^2 = 0.15$; number of cigarettes: $F(3, 192) = 11.20$, $p < 0.01$, $\eta_p^2 = 0.15$; quitting: $F(3, 189) = 7.48$, $p < 0.01$, $\eta_p^2 = 0.11$	There were significant decreases in nicotine dependence, number of cigarettes smoked and increases in quitting in the CP condition, relative to the control conditions	193	$d = 0.61$
Armitage and Arden (2012)*	Alcohol consumption (in units, i.e., eight grams alcohol per day) after the intervention: CP multiple plans: ($M = 2.33$, SD $= 1.93$); CP single: ($M = 3.41$, SD $= 2.02$); control: ($M = 3.96$, SD $= 3.54$)	Alcohol consumption was significantly higher in the active control condition compared with each of the experimental conditions	69	$d = 0.38$
Armitage and Arden (2010)*	After the intervention PA: $F(1, 66) = 7.28$, $p < 0.01$, $\eta_p^2 = 0.10$	There was significant increase in physical activity in the CP group but not in the control group	68	$d = 0.14$
Murgraff et al. (1996)*	The difference in the drinking frequency between two groups after the intervention: $X2 = 5.32$, df $= 1$; $p = 0.05$	There were significant decreases in alcohol drinking frequency in the CP group compared to the control group	102	$d = 0.68$
Rutter et al. (2006)	After intervention (ITT analysis): CP: 731 (78.94%); 195 (21.06%); C: 777 (80.26%); 191 (19.74%); OR $= 0.9214$	There was non-significant difference in uptake of mammography rates between CP condition and control condition (expressed as OR: number of people who attended screening relative to number of people who did not attend)	2082	
Skar et al. (2011)	After eight weeks: CP: ($M = 2.76$, SD $= 2.30$); control: ($M = 2.96$, SD $= 2.31$); $p > 0.05$	There was non-significant increase of physical activity reported as frequency (times per week) in CP group compared to the control group	1273	$d = 0.09$
van Osch et al. (2008)	After intervention (ITT analysis): point prevalence abstinence: CP: 146 (19.0%); C: 134 (16.6%); OR $= 1.21$; continuous abstinence: CP: 103 (13.4%); C: 85 (10.5%); OR $= 1.35$	There was non-significant difference in smoking cessation rates – reported as proportion [N (%)] and OR	1566	

Notes: Number (%) and odds ratios or statistical significance.
CP, coping plans; C, control condition; OR, odds ratio.
*There was a statistically significant difference between CP and control condition for $p < 0.05$. We prioritised data provided to us by study authors; thus data may differ from that reported in the published papers.To estimate the effect size of RCTs with control and active control condition we calculated the mean of control and active control conditions and compared it to the CP condition mean. To estimate the effect size of RCTs with single and multiple CPs condition we calculated the mean of single and multiple CPs conditions.

Three large trials comparing a separate coping planning condition against no intervention controls (involving a total of 4921 participants) showed that coping planning interventions alone were insufficient to enhance mammography uptake (Rutter et al., 2006), increase physical activity levels (Skar et al., 2011) or promote smoking cessation (van Osch et al., 2008). In the smoking cessation study, the authors reported a significant intervention effect of reducing smoking in the pre-protocol analyses; however, intention-to-treat (ITT) analysis revealed non-significant effects on point prevalence abstinence rates and on continuous refraining from smoking (van Osch et al., 2008).

Potential moderators of effects

Based on the limited number of included trials, available statistical power for further moderator analyses was limited. Five potential moderators were considered and descriptive results are: (1) the type of behaviour (i.e., uptake of health-promoting behaviour vs. reduction of health-impairing behaviour), (2) outcome measures (i.e., objective vs. self-reported outcome measures), (3) length of time between baseline and follow-up, (4) changes in self-efficacy and (5) changes in motivation/intention from baseline to follow-up (see Table 4). We found no evidence for the type of behaviour, outcome measures or the length of follow-up periods to moderate intervention effects. When changes in self-efficacy and intentions were measured as secondary trial outcomes, significant changes favouring the intervention group were found except for one study which found changes in self-efficacy but not in intention (Arbour-Nicitopoulos et al., 2009). In all cases, when significant changes in self-efficacy were found, the trial was efficacious. However, none of the non-significant trials measured self-efficacy as secondary outcome. It is therefore not possible to test whether significant changes in self-efficacy are a condition for successful behaviour change in coping planning interventions. Further research is needed to explore under which conditions coping planning interventions are efficacious in changing health-related behaviour.

Fidelity and adherence to experimental instructions to form plans

In some of the larger included trials, it was observed that not all participants allocated to the coping planning condition formed coping plans when prompted to do so or reported that planning was irrelevant to them (Rutter et al., 2006; Skar et al., 2011). When the results were separately analysed for respondents who adhered to the intervention protocol, participants in the intervention condition showed more behaviour change than controls (Rutter et al., 2006; Skar et al., 2011). The results of included studies in this review suggest that supporting participants in forming specific and relevant plans may have an impact on fidelity and thereby to the efficacy of health-related interventions.

Discussion

This is the first systematic review to investigate the efficacy of coping planning in health-related interventions, scrutinising randomised controlled trials in which participants were randomly allocated to conditions which varied only the use of

Table 4. Potential moderators of the effects.

Study	Type of behaviour (health promoting versus health impairing)	Outcome measures (objective versus self-reported)	Follow-up period (long: >eight weeks; short ≤ eight weeks)	Change in self-efficacy from baseline to follow-up [increased (+), or not changed (o)]	Changes in intention from baseline to follow-up [increased (+), or not changed (o)]
Trials with significant effects on behaviour					
Arbour-Nicitopoulos et al. (2009)	Promoting	Self-reported	Long	+	o
Arden and Armitage (2012)	Impairing	Self-reported	Short	Not reported	Not reported
Armitage (2008)	Impairing	Self-reported	Short	+	+
Armitage and Arden (2012)	Impairing	Self-reported	Long	+	+
Armitage and Arden (2010)	Promoting	Self-reported	Short	+	+
Murgraff et al. (1996)	Impairing	Self-reported	Short	Not reported	Not reported
Scholz et al. (2007)	Promoting	Self-reported	Long	Not reported	+
Sniehotta et al. (2006)	Promoting	Self-reported	Long	+	+
Trials without significant effects on behaviour					
Rutter et al. (2006)	Promoting	Objective	Long	Not reported	Not reported
Skar et al. (2011)	Promoting	Both	Short	Not reported	+
van Osch et al. (2008)	Impairing	Both	Long	Not reported	Not reported

coping plans. The overall efficacy of coping planning was variable. The results suggest that forming action plans *and* coping plans is more efficacious in promoting behaviour change than forming only action plans; demonstrated in four out of five comparisons. Four identified interventions testing coping plans against control using pre-specified plans for overcoming barriers were efficacious in changing health-related behaviour whilst the three interventions that did not provide pre-specified situational barriers failed to show a significant coping planning effect.

Coping planning used in practice

This systematic review included three large studies comparing coping planning against no intervention controls, each had no intervention effect across the participating sample, using intention to treat analyses (Rutter et al., 2006; Skar et al., 2011; (van Osch et al., 2008). There are two main similarities among those three interventions: (1) they were all delivered without expert assistance (two were delivered online and one was a postal questionnaire) and (2) formulated plans were not assessed and feedback on them was not given. The authors of studies that employed coping plans have claimed that these self-regulation interventions are of minimal intensity, easy to administer and highly time efficient (van Osch et al., 2008); however, the challenge faced by the intervention providers was to prompt participants to form meaningful and appropriate coping strategies.

In interventions delivered face-to-face, coping plan formulation was assisted by an intervener or structured materials and participants received guidance on how to form plans to overcome anticipated barriers and/or feedback about their plans. It seems plausible that the interaction with a facilitator during a behaviour change intervention including planning components influences efficacy of the intervention (Griffiths, Lindenmeyer, Powell, Lowe, & Thorogood, 2006). Whilst facilitator contact may moderate efficacy by ensuring that participants complete the intervention as intended, it may not be a necessary precondition, as evidence by the use of volitional help sheets (Arden & Armitage, 2012; Armitage, 2008; Armitage & Arden, 2010). Health-related interventions which include planning components can be easily delivered through email, phone calls or phone applications. However, making plans appears to be efficacious when plan formation is monitored, supervised or when plans are pre-specified. Similar results were observed in the study which measured long-term adherence to a physical activity intervention and compared telephone-assisted and self-administered coping plans (Evers, Klusmann, Ziegelmann, Schwarzer, & Heuser, 2011); study participants supported by telephone-assisted intervention reported significantly more coping plans than study participants supported by self-administered coping planning intervention. Plans can serve as an efficacious behaviour change technique in health-related consultations (e.g., as a technique that general practitioners can use to promote increases in physical activity). However, they do not appear to be equally efficacious in interventions which do not give any support in formulating plans.

Fidelity and adherence to experimental instructions to form plans

The results of the included studies suggest that one of the limitations of planning interventions is the lack of fidelity to the intervention protocol, especially in

interventions which are not delivered face-to-face. Participants often do not form meaningful plans when prompted to do so. Moreover, there is limited knowledge about what constitutes a good plan. Most of the included interventions reported that when the results were separately analysed for respondents who adhered to the intervention protocol, the intervention was more efficacious (van Osch et al., 2008). Similar results have been shown in other health-related studies (Michie, Dormandy, & Marteau, 2004). This review suggests that monitoring whether participants form relevant plans may increase the efficacy of the health-related interventions.

Implications for practice and future research

The results of this review show that despite the fact that numerous studies employed coping plans as a part of multi-component interventions, not many of them applied a design which would allow evaluation of the efficacy of coping planning only or the additional effect of adding coping planning to complex interventions. While complex interventions are important in promoting behaviour change, randomised trials disentangling various behaviour change techniques used in health-related interventions are needed to develop our knowledge of the efficacy of individual techniques.

Intervention descriptions should accurately specify theory used to design the intervention, behaviour change techniques and modes of delivery, so that effective interventions can be replicated (Davidson et al., 2003; Webb, Sniehotta, & Michie, 2010). We still lack a clear understanding of effective techniques and mechanisms of change across behaviours, populations and settings. Thus, designing trials which allow us to separate specific techniques of the interventions (such as coping plans) would let us evaluate the efficacy of various parts of interventions.

Limitations

Few interventions were designed in such a way to specifically investigate the contribution of coping planning. More interventions using factorial designs that allow testing main and interaction effects of action and coping planning are advocated, bearing in mind potential moderators of efficacy. Another potential limitation of this study is publication bias, which we attempted to minimise through calls for unpublished papers. However, we experienced difficulties obtaining unpublished data-sets referenced in the available literature.

Another important limitation is variability in construct labelling and imprecise reporting of RCTs. Coping planning in the health psychology literature has been labelled as: ‘reasoning implementation intentions’ (Prestwich, Ayres, & Lawton, 2008), ‘shielding goal striving from unwanted thoughts and feelings’ (Achtziger, Gollwitzer, & Sheeran, 2008), ‘shielding goals from unwanted responses’ (Gollwitzer & Sheeran, 2006) or descriptively ‘planning how to overcome behaviour change barriers’. There is a need for a more consistent use of terminology to describe intervention techniques (Dombrowski, Sniehotta, Avenell, & Coyne, 2007), and when new terms are coined, that their distinctions be explicitly contrasted against existing techniques that may be similar.

Conclusions

Combining action plans with coping plans in health-related behaviour change interventions is more efficacious than using action plans only, though prompting the formulation of only coping plans appears to be insufficient to successfully change participants' health-related behaviours when plans are not pre-specified. There is a need for more high quality RCTs testing the efficacy and effect modifiers of coping planning as a technique for health behaviour change.

Acknowledgements

Martin White is funded in part, and Dominika Kwasnicka and Falko F. Sniehotta are funded in full by Fuse, the Centre for Translational Research in Public Health, a UKCRC Public Health Research Centre of Excellence. Funding for Fuse from the British Heart Foundation, Cancer Research UK, Economic and Social Research Council, Medical Research Council, the National Institute for Health Research, under the auspices of the UK Clinical Research Collaboration, is gratefully acknowledged.

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